

Project README

Design and Simulation of TPS Worldwide's Enterprise Network Infrastructure

Overview

This project showcases a comprehensive network infrastructure simulation for **TPS Worldwide**, implemented using Cisco Packet Tracer. The design focuses on scalability, security, and the logical segmentation of a multi-department fintech organization.

Key Network Features

- **Two-Tier Topology:** Implementation of Core and Access layers for optimized traffic flow.
- **Logical Segmentation:** VLANs (Development, HR, Admin, etc.) used to isolate traffic and improve security.
- **Centralized Infrastructure:** Dedicated VLAN for DHCP, DNS, FTP, and Web servers.
- **Inter-VLAN Routing:** Layer 3 switching enables controlled communication across the organization.

Technical Configurations

- **VLAN 10:** Development
- **VLAN 30:** HR
- **VLAN 50:** Server Infrastructure (192.168.50.0/24)
- **VLAN 70:** Guest Network
- **Services:** DNS (A-records), DHCP (Dynamic pools), FTP (Secure transfers).

Validation & Results

The network was validated through end-to-end connectivity tests, ensuring that all departments can access necessary resources while maintaining departmental isolation where required.

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